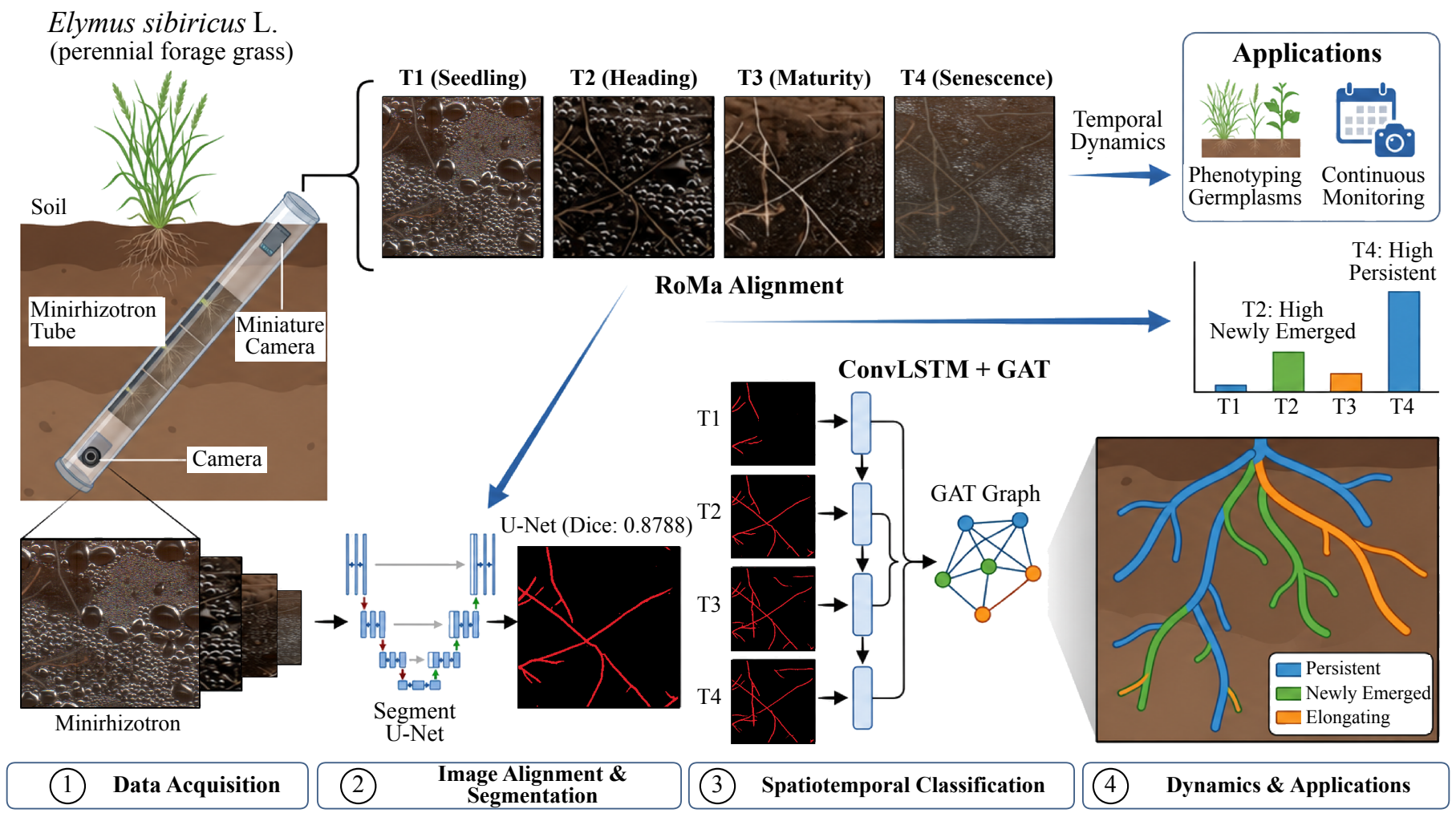

Elymus sibiricus L.
(perennial forage grass)
Soil
Minirhizotron Tube
Miniature Camera
Camera
Minirhizotron
T1 (Seedling)
T2 (Heading)
T3 (Maturity)
T4 (Senescence)
Temporal Dynamics
Applications
Phenotyping Germplasms
Continuous Monitoring
RoMa Alignment
T4: High Persistent
T2: High Newly Emerged
T1
T2
T3
T4
ConvLSTM + GAT
GAT Graph
U-Net (Dice: 0.8788)
Segment U-Net
Persistent
Newly Emerged
Elongating
1 Data Acquisition
2 Image Alignment & Segmentation
3 Spatiotemporal Classification
4 Dynamics & Applications